

HAT-P-54b

R-1187

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_220221 · created 2026-07-31T05:26:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459631.7755 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.053
Transit depth (Rp/Rs)^2	2.03 +/- 1.5 [%]
Orbital Inclination (inc)	87.81 +/- 2.16
Airmass coefficient 1 (a1)	0.874 +/- 0.035
Airmass coefficient 2 (a2)	0.111 +/- 0.035
Scatter in the residuals of the lightcurve fit is	4.04 %
Best Comparison Star	#2 - [188, 359]
Optimal Aperture	3.02
Optimal Annulus	12.94
Transit Duration (day)	0.0669 +/- 0.0087

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.142 $\pm$ 0.053 vs 0.1572 (deviation 0.29 σ)
Duration fit vs published	0.0669 d vs 0.075 d (deviation 0.93 σ)
Mid-time O-C	-0.6 min
Depth SNR	2.7
Residual scatter	4.04 %

Light curve

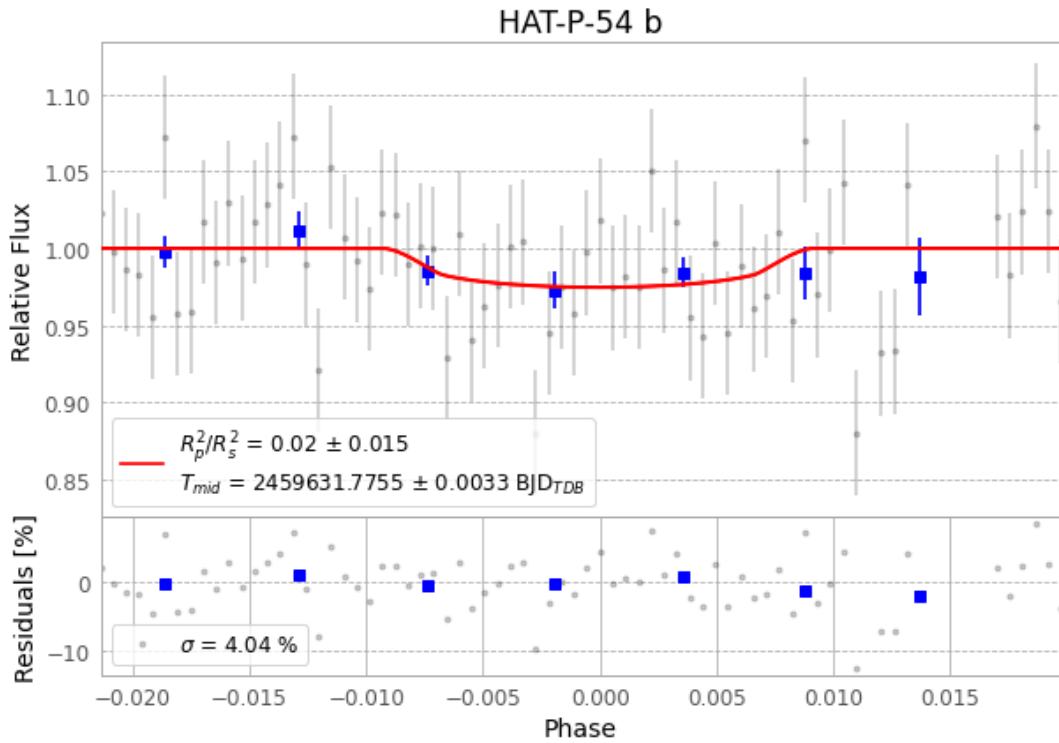


Figure 1 — FinalLightCurve_HAT-P-54 b_2022-02-20.png (SHA-256 9c4334797c1020f4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 188], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ff4d7f7f0c3e4b72...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), HATP-54220221043312.FITS → HATP-54220221081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-220221.log (3fa81b65d389...), FinalLightCurve_HAT-P-54 b_2022-02-20.png (9c4334797c10...), FinalLightCurve_HAT-P-54 b_2022-02-20.csv (937fba8cedae...)

Compute resources

Wall time 425.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:26Z.